Synopsis of Introductory Case: Coca-Cola and PepsiCo Stock Returns

This case examines the performance and risk characteristics of the weekly stock returns of **The Coca-Cola Company** and **PepsiCo** using descriptive statistics and regression analysis. Because both companies operate in the same beverage industry and compete in similar markets, comparing their returns provides useful insights into their risk, variability, and potential relationship.

The descriptive statistics show differences in both return levels and volatility. The **mean weekly return** for Coca-Cola is **0.32%**, which is slightly higher than PepsiCo's **0.20%**. However, PepsiCo exhibits greater variability in its returns. The **standard deviation** for PepsiCo is **3.16%**, compared with **1.94% for Coca-Cola**, indicating that PepsiCo's returns fluctuate more widely around the average. This higher variability is also reflected in the **range of returns**, which is **17.60% for PepsiCo** compared with **11.20% for Coca-Cola**.

Examining the distribution of returns provides additional insights. The **interquartile range (IQR)** for Coca-Cola is **3.06%**, while PepsiCo's IQR is **3.39%**, suggesting slightly greater dispersion in PepsiCo's middle 50% of observations.

Risk relative to return can be evaluated using the **coefficient of variation**. Coca-Cola has a coefficient of variation of **6.11**, while PepsiCo's coefficient of variation is **16.05**, indicating that PepsiCo carries significantly more risk per unit of expected return. Similarly, the **mean absolute deviation** is higher for PepsiCo (**2.49%**) than for Coca-Cola (**1.57%**), further confirming that PepsiCo's returns vary more from their average value.

The **Sharpe ratio**, which measures risk-adjusted performance, is **–1.88 for Coca-Cola** and **–1.19 for PepsiCo**. Negative values indicate that both stocks performed worse than the risk-free rate during the analyzed period. However, PepsiCo's Sharpe ratio is less negative, suggesting slightly better performance relative to risk.

To examine whether the two stocks move together, a regression analysis was conducted with PepsiCo returns as the dependent variable and Coca-Cola returns as the independent variable.

The **R^2 value of 0.304** indicates that approximately **30.4% of the variation in PepsiCo's returns can be explained by movements in Coca-Cola's returns**. The coefficient for Coca-Cola returns is positive and statistically significant, suggesting that when Coca-Cola's returns increase, PepsiCo's returns tend to increase as well.

In summary, the analysis shows that **Coca-Cola provides slightly higher average returns with lower volatility**, while **PepsiCo exhibits greater variability and higher relative risk**. The regression results further indicate a **moderate positive relationship between the returns of the two companies**, reflecting their similar industry exposure and market conditions.